FIITJEE REVIEW TEST-05

(BATCHES: 921 LOT)

IIT – JEE 2021

Time: 3 Hours **Maximum Marks: 177**

A. Question Paper Format

1. The question paper consists of 3 sections (Physics-**Section-I**, Chemistry-**Section-II,** Mathematics-**Section-III**) and each part consists of four parts.
2. **Part A – (I)** contains **4** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part A – (II)** contains **5** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part A – (III)** contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. . Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-B (01-05)** contains five (05) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

B. Marking scheme:

5. For each question in Part A (I), you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.
6. For each question in Part A (II), you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.
   Partial Marks: **+1** For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
   Zero Marks: **0** If none of the bubbles is darkened.
   ***Negative Marks*: −2 In all other cases.**
   For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **−2** marks, as a wrong option is also darkened.
7. For each question in Part A (III), you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.
8. For each question in Part B, you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **Date:** . . . . . .

SECTION - I: PHYSICS

PART – A(I) (Single Correct Choice Type)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. The cylindrical tube of a spray pump has a radius R, one end of which has n fine holes, each of radius r. If the speed of flow of the liquid in the tube is V, the speed of ejection of the liquid through the holes is

(A) $V \times \frac{R}{r}$ (B) $V \times \left(\frac{r}{R}\right)^2 \times n$

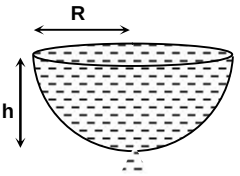
(C) $\frac{V}{n} \times \left(\frac{R}{r}\right)^2$ (D) $\frac{V}{n} \times \frac{R}{r}$

2. Water rises to a height of 10cm in a capillary tube and mercury falls to a depth of 3.42 cm in the same capillary tube. If the density of mercury is 13.6g/cc and the angle of contact of mercury and water are 135° and 0°, respectively, the ratio of surface tension of water and mercury is

(A) 1 : 0.15 (B) 1 : 3

(C) 1 : 6.5 (D) 1.5 : 1

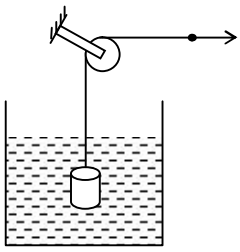
3. A hemispherical tank of radius 'R' is completely filled with water. Now an orifice of small area 'a' is made at the bottom of tank. The time required to empty the tank is

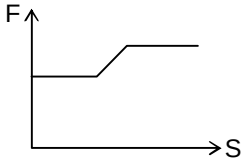


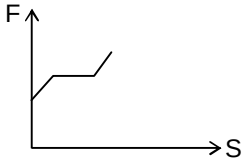
(A) $\frac{14\pi R^{5/2}}{15\sqrt{2}\, a\sqrt{g}}$ (B) $\frac{14\pi R^{5/2}}{\sqrt{2}\, a\sqrt{g}}$

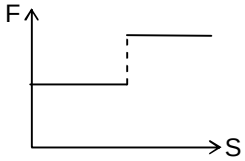
(C) $\frac{14\pi R^{5/2}}{5\sqrt{2}\, a\sqrt{g}}$ (D) $\frac{4\pi R^{5/2}}{15\sqrt{2}\, a\sqrt{g}}$

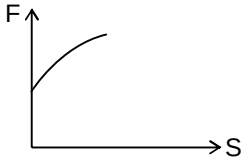
4. A metallic cylinder of radius r and height h is totally submerged in a liquid contained in a beaker as shown in the figure. If force F is pulling the string at a constant velocity v, the graph showing the variation of force F against displacement of cylinder is best represented by



(A) 

(B) 

(C) 

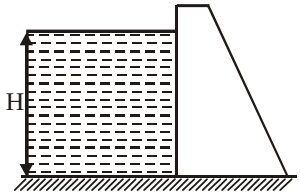
(D) 

Space for rough work

PART – A(II) (Multiple Correct Choice Type)

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

5. A wall of length ℓ supports water to a height H as shown in figure. Choose the correct statement(s). Take ρ as the density of water.



(A) the force exerted by water on the wall per unit length is $\frac{1}{2}\rho g H^2$

(B) the force exerted by water on the wall is $\frac{1}{2}\rho g H^2 \ell$

(C) the point of application of the resultant force acts $\frac{2}{3}H$ below the free surface

(D) the point of application of the resultant force acts $\frac{H}{3}$ below the free surface

6. Water flows steadily through a horizontal pipe of variable cross-section. If the pressure of water is ρ at a point where the velocity of flow is v, at another point (pressure p′), where the velocity of flow in ηv: The following statements are given below

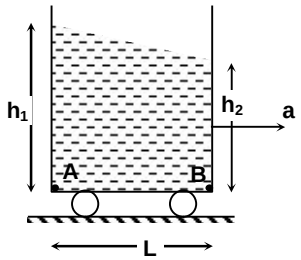
(A) $\frac{p'}{p} < 1$, when $\eta < 1$

(B) $\frac{p'}{p} < 1$, when $\eta > 1$

(C) $\frac{p'}{p} > 1$, when $\eta < 1$

(D) $\frac{p'}{p} > 1$, when $\eta > 1$

7. A beaker is accelerated uniformly as shown. A and B are two points at the bottom and h_1, h_2 are the corresponding depths from the top. [ρ_A = pressure on A, p_0 = pressure at B, ρ = density of the fluid]. Choose the correct option.



(A) $p_A > p_B$
(B) $p_A = p_B$
(C) $p_A < p_B$
(D) $\rho g h_1 - \rho a L = \rho g h_2$

8. A small block of wood, of specific gravity 0.4, is submerged in water at a depth of 2.9 m. If the block is released from rest. Then choose the correct alternative(s).
(A) Acceleration of block towards the surface is 14.7 m/s^2
(B) Velocity of block after 2 sec is 29.4 m/s.
(C) Time taken by block to reach the surface is 0.63 sec.
(D) Time taken by block to reach the surface is 2.74 sec.

9. A cubical block is floating in a liquid with one third of it's volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2.
(A) The fraction of volume immersed in the liquid will change.
(B) The buoyancy force on the block will change.
(C) The buoyancy force will increase by 50 percent.
(D) The pressure in the liquid will increased.

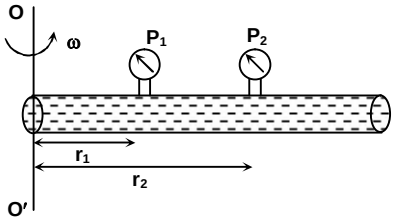
Space for rough work

PART – A: (Paragraph Type)

*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 10 to 11

A tube filled with water (density ρ_w) and closed at both ends uniformly rotates in a horizontal plane about OO′-axis. The manometers fixed in the tube wall at distance r_1 and r_2 from the rotational axis indicate pressure P_1 and P_2 respectively as shown in the figure. Then, answer the following questions.



10. Angular velocity 'ω' of rotation of the tube.

(A) $2\sqrt{\dfrac{2(P_1-P_2)}{\rho_w(r_1^2-r_2^2)}}$ (B) $\sqrt{\dfrac{(P_1-P_2)}{\rho_w(r_1^2-r_2^2)}}$

(C) $\sqrt{\dfrac{(P_1-P_2)}{2\rho_w(r_1^2-r_2^2)}}$ (D) $\sqrt{\dfrac{2(P_1-P_2)}{\rho_w(r_1^2-r_2^2)}}$

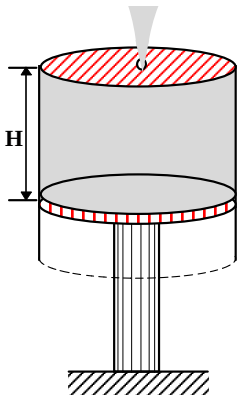
11. Pressure inside the water touching the axis OO′.

(A) $\dfrac{P_2r_1^2-P_1r_2^2}{2(r_1^2-r_2^2)}$ (B) $\dfrac{P_2r_1^2+P_1r_2^2}{(r_1^2+r_2^2)}$

(C) $\dfrac{P_2r_1^2-P_1r_2^2}{(r_1^2-r_2^2)}$ (D) $\dfrac{P_2r_1^2+P_1r_2^2}{2(r_1^2+r_2^2)}$

Space for rough work

Paragraph for Question Nos. 12 to 13



In the adjacent figure a cylindrical vessel of mass M and cross-sectional area A is placed inverted on a fixed smooth piston of same cross-sectional area fixed to the ground. The space between the cylinder and piston is completely filled with liquid of density ρ. There is a small orifice of cross-sectional area a. ($a \ll A$) at the top flat portion of this vessel. Now answer the following questions. (initially length of the liquid column in the vessel is H)

12. Speed of the liquid with which it comes out of the vessel is

(A) $\sqrt{2gH}$

(B) $\sqrt{\frac{2Mg}{\rho A}}$

(C) $\sqrt{\frac{Mg}{2\rho A}}$

(D) $\sqrt{\frac{Mg}{\rho A}}$

13. The time in which the total liquid comes out of the vessel is

(A) $\frac{H}{a}\sqrt{\frac{\rho A^3}{2Mg}}$

(B) $\frac{H}{a}\sqrt{\frac{\rho A^3}{Mg}}$

(C) $\frac{H}{a}\sqrt{\frac{2\rho A^3}{Mg}}$

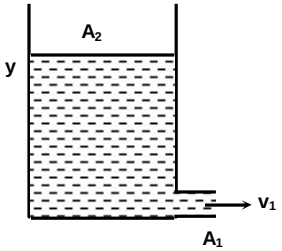
(D) $\frac{H}{a}\sqrt{\frac{A^3}{2gH}}$

Space for rough work

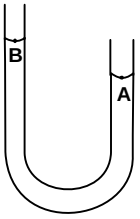
SECTION – C
(Numerical Answer Type)

This section contains **05** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

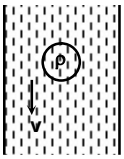
1. A fixed container having area of cross-section 1 m^2 contains water. The initial height of Water in container is 1m. The container has an orifice of small cross section area near its bottom. Find the ratio of R_1/R_2, where R_1 and R_2 are the rates of change of height of the liquid in the container when the heights of liquid in container are 1 m and 0.25 m respectively. Area of orifice is 0.001 m^2.



2. The difference in water levels in the two communicating capillary tubes of different diameter d = 1 mm and d = 1.5 mm is K × 2.38. Surface tension of water = 0.07 N/m and angle of contact between glass and water is 0°. Find the value of K.

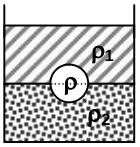


3. A liquid of density ρ = 1000 kg/m^3 viscosity η = 0.1 Ns/m^2 is flowing down in vertical pipe of large cross section. A small ball of density ρ_0 = 100 kg/m^3 and r = 5 cm will be at rest in flowing liquid, if velocity of flowing liquid is 10k m/s. Then find the value of k.



4. When a tangential force of 0.02 N is applied on a large wooden plate of area 10 m^2 floating on the surface of river, plate moves with the speed 2 m/s on the river surface. The river is 1 m deep and the water in contact with the bed is stationary. Assuming constant speed gradient, coefficient of viscosity of water is 10^{-N} poise. Find the value of N

5. A solid uniform ball having volume V and density ρ floats at the interface of two immiscible liquids as shown in the figure. The densities of the upper and the lower liquids are ρ_1 and ρ_2 respectively such $\rho_1 < \rho < \rho_2$. Find the ratio of the volume of the ball in the upper liquid to the lower liquid? ($\rho_1 = \rho/2$, $\rho_2 = 5\rho$)



Space for rough work

FIITJEE REVIEW TEST-05

(BATCHES: 921 LOT)

IIT – JEE 2021

ANSWERS

PHYSICS [SECTION-I]

1.	**C**	2.	**C**	3.	**A**	4.	**A**
5.	**A, B, C**	6.	**B, C**	7.	**A, D**	8.	**A, C**
9.	**B, C, D**	10.	**D**	11.	**C**	12.	**B**
13.	**C**	1.	**2**	2.	**2**	3.	**5**
4.	**2**	5.	**8**				

Solutions

1. From continuity equation, $A_1V_1 = A_2V_2$

$$\Rightarrow v = \left(\frac{R}{r}\right)^2 \times \frac{V}{n}$$

2. $$h = \frac{2T\ \cos\theta}{r\rho g}$$

or $$s = \frac{hr\rho g}{2\cos\theta}$$

or $$S \propto \frac{h\rho}{\cos\theta}$$

$$\frac{S_W}{S_{Hg}} = \frac{h_1}{h_2} \times \frac{\cos\theta_2}{\cos\theta_1} \times \frac{\rho_1}{\rho_2}$$

Putting the values, we obtain 1 : 6.5

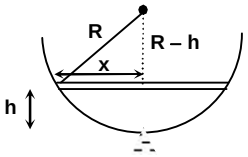
3. $dV = \pi x^2 dh$

$= \pi(2Rh - h^2)dh$

$$-\frac{dV}{dt} = a\sqrt{2gh}$$

$$\int dt = \int \frac{-\pi}{a\sqrt{2g}}(2Rh^{1/2} - h^{3/2})dh$$

$$t = \frac{14\pi R^{5/2}}{15\sqrt{2}\ a\sqrt{g}}$$

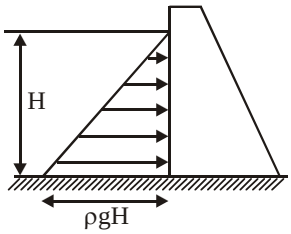


4. As long as cylinder is completely inside, B is constant in magnitude, then its value decreases linearly with displacement and finally becomes zero.

5. The pressure at the base of the dam is ρgH. The force per unit length of the dam is equal to the area of the pressure diagram

$$\frac{F}{\ell} = \frac{1}{2}(\rho gH)H = \frac{1}{2}\rho gH^2$$

or $$F = \frac{1}{2}\rho gH^2\ell$$



The point of application of the resultant force passes through the centroid of the pressure diagram, i.e., at $\frac{H}{3}$ from the base or $\frac{2}{3}H$ from the top.

6. $\eta > 1$, velocity at another point will be greater then v
so, pressure (p′) < p
$\eta < 1$, velocity at another point will be less then v
so, pressure (p′) > p

8. $$a = \left(\frac{\rho_W}{\rho_B} - 1\right)g = 14.7\ \text{m/s}^2$$

$$t = \sqrt{\frac{2s}{a}} = 0.63\ \text{sec}$$

9. If the volume immersed initially is (V/3). Then $\frac{V}{3}\rho g = mg$ ...(i)

If the volume immersed when the system accelerates is V′ then

$$V'\rho\left(g+\frac{g}{2}\right)-mg=\frac{mg}{2}$$

$$\Rightarrow V'=\frac{V}{3}$$

13. Applying Bernoulli's equation at the points 1 and 2 , we get

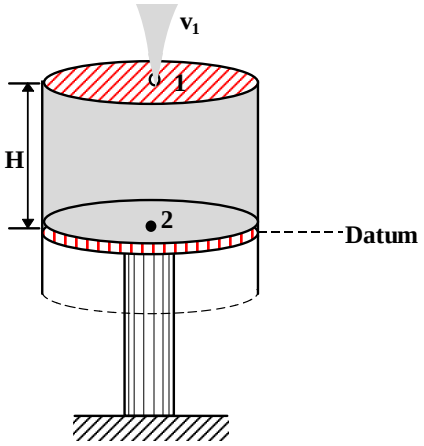
$$\frac{p_1}{\rho g}+\frac{v_1^2}{2g}+H=\frac{p_2}{\rho g}+\frac{v_2^2}{2g}+0$$

Here $p_1 = p_0$, $p_2 = p_0 + \rho gH + \frac{Mg}{A}$

$$\frac{v_2^2}{2g}\approx 0 \quad (a << A)$$

$$\therefore \quad v_1 = \sqrt{\frac{2Mg}{\rho A}}$$

Using continuity equation , $v_2 = \frac{av_1}{A}$



1. $v = \sqrt{2gh}$

Change in height of the liquid

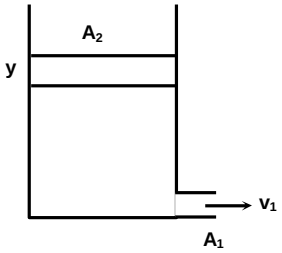
$$R = \frac{dy}{dt}=\left(\frac{A_1}{A_2}\right)v_1$$

Now, $R_1=\left(\frac{0.001}{1}\right)\times\sqrt{2\times10\times1}$ . . . (1)

and $R_2=\left(\frac{0.001}{1}\right)\times\sqrt{2\times10\times0.25}$ . . . (2)

$\therefore$ from (1) and (2)

$$\frac{R_1}{R_2}=\sqrt{\frac{1}{0.25}}=2$$



2. $2T\cos\theta\left(\frac{1}{r_1}-\frac{1}{r_2}\right)=\rho gh$

3. In fluid frame ball is moving with constant velocity (terminal) and resultant of all forces acting on it zero.

(A) Buoyant force

(B) viscous force

(C) mg

$$v_t=\frac{2}{9}\frac{(\rho-\rho_0)gr^2}{\eta}=\frac{2}{9}\times\frac{900\times10}{0.1}\times25\times10^{-4}=50\ \text{m/s}$$

4. $$n=\frac{\left(\frac{F}{A}\right)}{\left(\frac{dv}{dx}\right)}=\frac{\frac{0.02}{10}}{\frac{2}{1}}=10^{-3}\text{kg/ms}=10^{-2}\text{poise}$$

5. $$\frac{V_1}{V_2}=\frac{\rho_2-\rho}{\rho-\rho_1}=8$$

FIITJEE REVIEW TEST-05

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

Time: 1 Hour **Maximum Marks: 59**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-04)** contains **04 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (05-09)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-A – (10-13)** contains **02 paragraphs**. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-C – (14-18)** contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks: – 2 In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part A (4),** you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **.Session :**. . . . . **Date:**. . . . . .

PART – A(I): (Single Correct Answer Type)

*This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

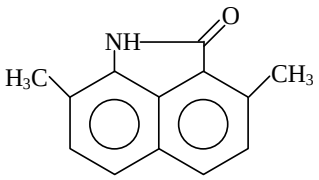
1. Deuterioethane on chlorination reacts as follows:

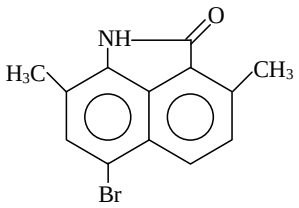
$$C_2H_5{-}D + Cl_2 \longrightarrow \underbrace{C_2H_5{-}Cl + D{-}Cl}_{7\%} + \underbrace{C_2H_4D{-}Cl + HCl}_{93\%}$$

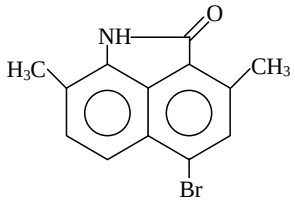
What is the reactivity ratio of H : D?

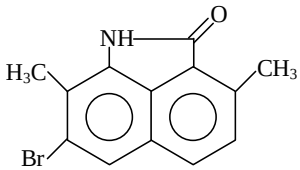
(A) 1 : 1 (B) 93 : 7 (C) 6 : 1 (D) 93 : 35

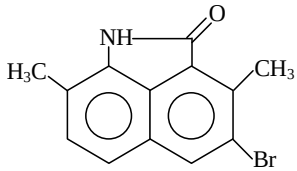
2. The major product obtained when Br_2/Fe is treated with



(A)


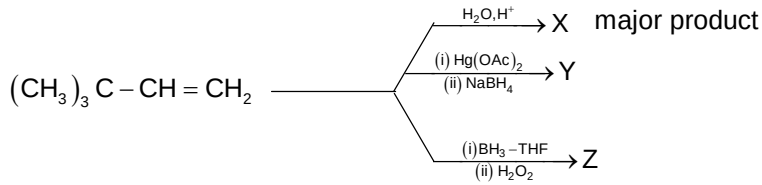
(B)


(C)


(D)


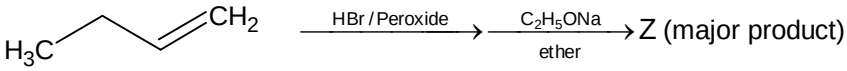
Space for rough work

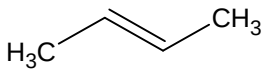
3.

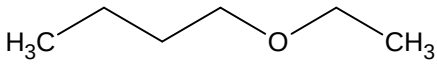


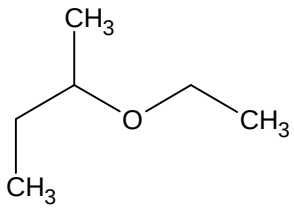
The correct statement from the above sequence is:
(A) X and Y are optically active where as Z is inactive
(B) Y and Z are optically active where as X is inactive
(C) X and Z are optically inactive where as Y is active
(D) Y and Z are optically inactive where as X is active

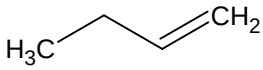
4. Identify Z in the following reaction sequence:



(A) 

(B) 

(C) 

(D) 

Space for rough work

PART – A(II): (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

5. CIS-2-butene $\xrightarrow[(ii)\ H_3O^+]{(i)\ HCO_3H}$ I

trans-2-butene $\xrightarrow[(ii)\ H_3O^+]{(i)\ HCO_3H}$ II

Correct statements are
(A) I is racemic mixture by anti addition
(B) II is meso compound by anti addition
(C) I is meso compound by syn addition
(D) II is racemic compound by syn addition

6. When nitrobenzene is treated with Br_2 in presence of $FeBr_3$, the major product formed is m–bromonitrobenzene. Statements which are related to obtain the m–isomer are
(A) the electron density on meta carbon is more than that on ortho and para positions.
(B) the intermediate carbonium ion formed after initial attack of Br^+ at the meta position is least destabilized.
(C) loss of aromaticity when Br^+ attacks at the ortho and para positions and not at meta position.
(D) None of these.

7. Which of the following carbides yield CH_4 gas on hydrolysis?

(A) Al_4C_3 (B) Be_2C (C) Mg_2C_3 (D) Na_2C_2

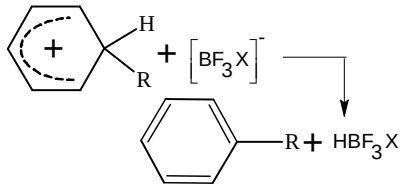
Space for rough work

8. Among the following the correct statements is/are
(A) The decreasing order of heat of combustion values is
cyclohexane > cyclopentane > cyclobutane > cyclopropane
(B) Cycloalkanes are planar
(C) Cyclopropane has higher heat of combustion per methylene group (CH_2) than that of cyclobutane
(D) With the exception of cyclopropane cycloalkanes are non-planar.

9. Alkylation of aromatic compounds can be done using BF_3. The mechanism of the reaction is as below:

$$RX + BF_3 \longrightarrow \left[\overset{\delta+}{R}.....X.....BF_3^{\delta-} \right] \longrightarrow \left[R^+\right]\left[BF_3X\right]^-$$

$$C_6H_6 + \left[R^+\right]\left[BF_3X\right]^- \rightarrow$$



In this reaction indentify the correct statements from the following codes.
(A) BF_3 acts as a Lewis acid
(B) BF_3 is used as a catalyst
(C) Amount of BF_3 used in the reaction is almost equal to amount of benzene used.
(D) BF_3 acts as a nucleophile

Space for rough work

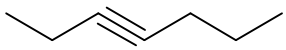
PART – A(III): (Paragraph Type)

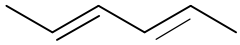
*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

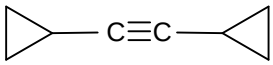
Paragraph for Question Nos. 10 to 11

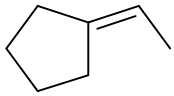
A hydrocarbon A, of the formula C_8H_{10}, on ozonolysis gives compound B($C_4H_6O_2$) only. The compound B can also be obtained from the alkyl bromide, C(C_3H_5Br) upon treatment with magnesium in dry ether, followed by carbon dioxide and acidification.

10. Compound A is

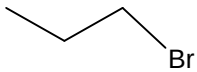
(A) 

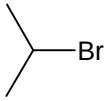
(B) 

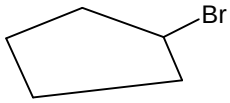
(C) 

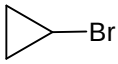
(D) 

11. Compound C is

(A) 

(B) 

(C) 

(D) 

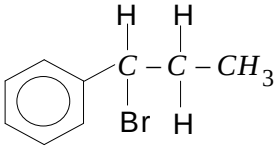
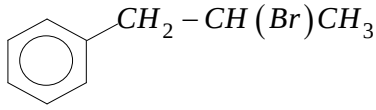
Space for rough work

Paragraph for Question Nos. 12 to 13

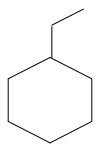
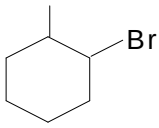
Dehydrobromination of (A) & (B) gives same alkene (C). Alkene (C) can be converted to (A) & (B) by addition of HBr in presence & absence of peroxide respectively. 1, 1 – diphenyl ethane is obtained on reaction of (C) with benzene in the presence of H^+ ions.

12. Compound 'A' & 'B' are

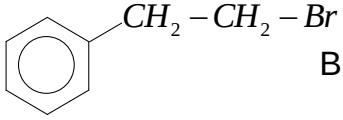
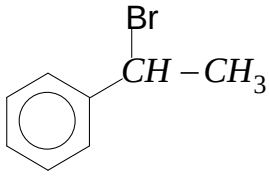
A)

A :  B : 

B)

A :  B : 

C)

A :  B : 

(D) None

13. A when treated with NBS & then with excess $NaNH_2$ gives (X), when (X) is reacted with $HgSO_4$ + dil. H_2SO_4 then it gives :-

(A) Phenol (B) Acetophenone

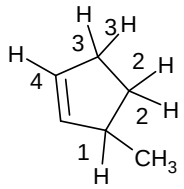
(C) Benzoic acid (D) None of these

Space for rough work

PART-C (Numerical Based Type)

This section contains 5 questions. The answer to each question is a single-digit integer, ranging from 0 to 9.

14. Which of the following C–H bond can give homolytic fission most easily



15. C_xH_y (organic compound, A) $\xrightarrow[\text{(Monobromination)}]{Br_2,h\nu}$ Product(Enantiomeric)

$A \xrightarrow{Br_2,CCl_4}$ no reaction

$A \xrightarrow{Br_2,Fe}$ mono bromo derivative product

What can be the minimum number of carbon atom (x) for the organic compound A?

16. In the monochlorination of 3- methylpentane, Let x be the number of pairs of isomers, which exist as enantiomers, y be the number of pairs of isomers, which exist as diastereomers, z be the number of isomers, which are achiral. Calculate the value of (x + y + z).

17. An unsaturated hydrocarbon X ($C_{14}H_{18}$) on complete hydrogenation give Y($C_{14}H_{26}$). X on reductive ozonolysis gives only two different products. None of the product contains more than two oxygen atoms per molecule. What will be the sum of position statements given below before statement, which may be applicable to compound X?
    (1) Double bond equivalent is six
    (2) Only one ring is present in the compound (X)
    (3) Two rings are present in the compound (X)
    (4) Triple bond may be present in the compound (X)

18. A 10 g mixture of isobutane and isobutene requires 20 g of Br_2 (in CCl_4) for complete addition. If 10g of the mixture is catalytically hydrogenated, 'a' gm of alkane is formed.
Find the value of (a-0.25)/5
(Atomic weight of bromine = 80).

Space for rough work

FIITJEE REVIEW TEST-05

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

ANSWER KEYS

1. **D**

2. **A**

3. **C**

4. **B**

5. **A, B**

6. **A**

7. **A, B**

8. **A, C, D**

9. **A, C**

10. **C**

11. **D**

12. **C**

13. **B**

14. **1**

15. **8**

16. **9**

17. **4**

18. **2**

SOLUTIONS

1. **D**

$C_2H_5–D + Cl_2 \longrightarrow \underbrace{C_2H_5–Cl + D–Cl}_{A\ (7\%)} + \underbrace{C_2H_4D–Cl + H–Cl}_{B\ (93\%)}$

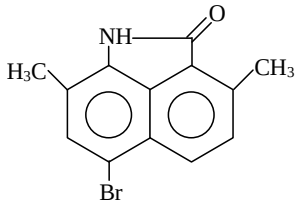
$$\frac{\text{Weight of (A)}}{\text{Weight of (B)}} = \left(\frac{\text{Reactivity of D}}{\text{Reactivity of H}}\right) \times \left(\frac{\text{Probability of D}}{\text{Probability of H}}\right)$$

$$\frac{7}{93} = \left(\frac{\text{Reactivity of D}}{\text{Reactivity of H}}\right) \times \frac{1}{5}$$

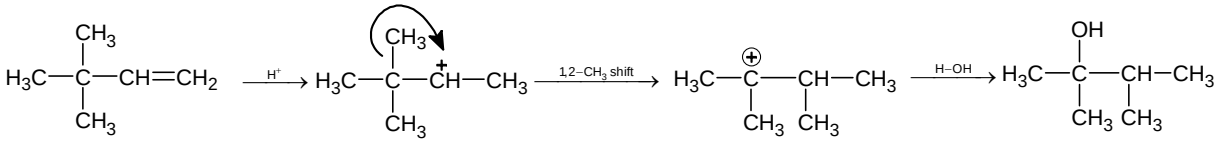
$$\left(\frac{\text{Reactivity of H}}{\text{Reactivity of D}}\right) = \frac{93}{35}.$$

2. **A**

Since first ring is activated. So $Br^{\oplus}$ will attack to the para position of the first ring with respect to NH.

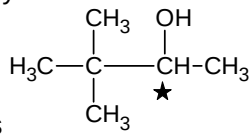


3. **C**

(A) 

Optically inactive (X)

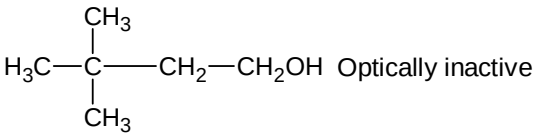
(B) On oxymercuration demercuration the product is alcohol without rearrangement



Optically active

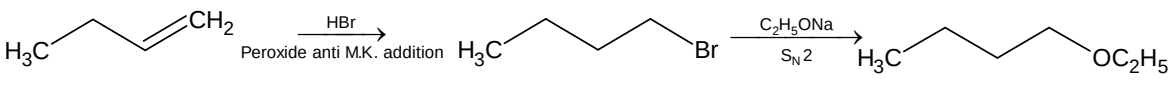
$\therefore$ Y is

(C) On hydroboration – oxidation the addition is against to Markonikov's rule



$\therefore$ Z is

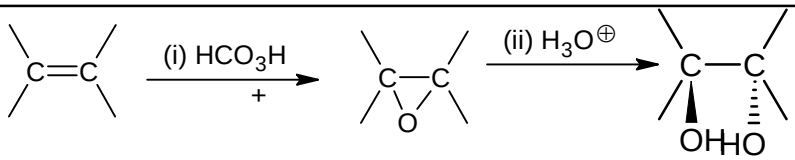
4. **B**



5. **A, B**

With HCO_3H followed by hydrolysis, it is anti addition

So overall reaction is anti addition of two OH group.



epoxy

anti addition

6. **A**

In nitrobenzene, $-NO_2$ group decreases electron density on ortho para position as compare to meta position.

7. **A, B**

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

8. **A, C, D**

Higher the number of carbon atoms, higher is the heat of combustion.
Cycloalkane are non- planer except cyclopropane (bond angle 60)

9. **A, C**

Since BF_3 is not finally regenerated it is not used as a catalyst and BF_3 is an electrophile not a nucleophile.

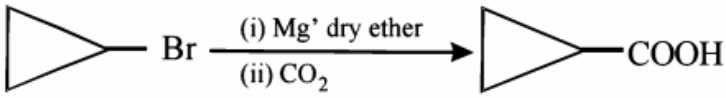
10. **C**

11. **D**

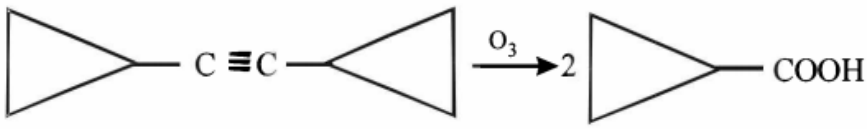
Solution:

$$\underset{(A)}{C_8H_{10}} \xrightarrow[(ii)\text{hydrolysis}]{(i)O_3} \underset{(B)}{C_4H_6O_2} \xleftarrow[(ii)CO_2,\,(iii)H^+]{(i)\text{Mg in dry ether}} \underset{(C)}{C_3H_5Br}$$

since compound (B) is obtained from compound (C) C_3H_5 Br through reaction with Mg and CO_2 , it seems that compound (B) is a carboxylic acid formed via the formation of Grignard reagent. Hence compound (C) should be an alkyl halide having three carbon atoms. The alkyl halide (C) is unsaturated (indicated by number of hydrogen atoms) which is present in the form of ring and thus (C) should be bromocyclopropane.



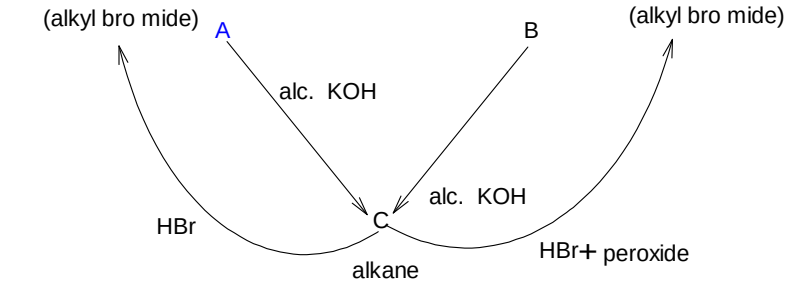
(C) C_3H_5Br (B) $C_4H_6O_2$

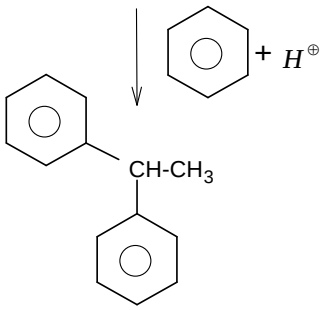


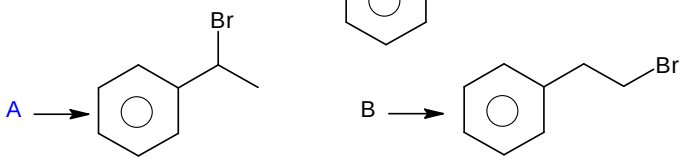
(A) C_8H_{10} (B) $C_4H_6O_2$

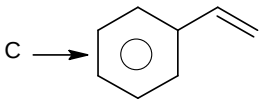
12. **C**

13. **B**







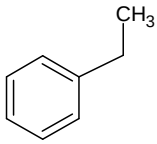


14. **1**

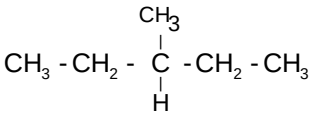
Stability of free radicls

Vinylic < 2° < allylic < substituted allylic

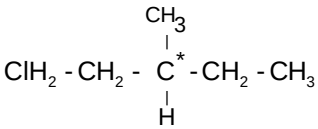
15. **8**



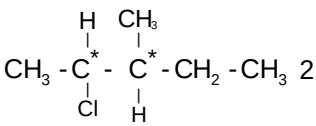
16. **9**



3- methyl pentane monochlorination, different products are

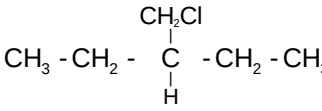


chiral, one chiral carbon atom pair of enantiomers

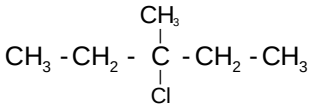


2 chiral carbon atoms 4 isomers. 2 pairs of enantiomers

4 pairs of diastereomers.



achiral molecule


achiral molecule

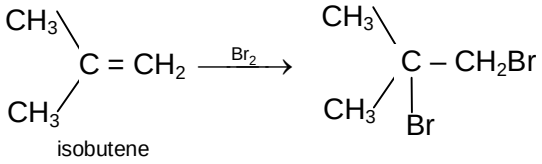
X = number of pairs of enantiomers = 3
Y = number of pairs of diastereomers = 4
Z = number of isomers which are achiral = 2
Total x + y + z = 9

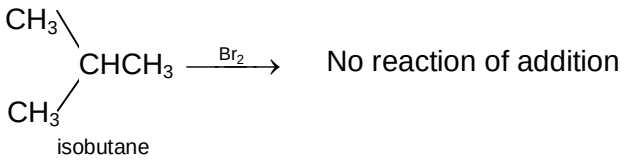
17. **4**

As on hydrogenation only 8 hydrogens are added so definitely two rings must be present to satisfy 6 DBE.
Triple bond is not present otherwise on reductive ozonolysis only two compounds with only two oxygen would not have formed.

18. **2**

Let isobutane by a g and isobutene be b g. Thus a + b = 10


isobutene


isobutane

Now ∴ 160g Br_2 is needed for 56 g isobutene

∴ 20g Br_2 is needed for $\frac{56 \times 20}{160}$ = 7g

Thus isobutene = 7g
Isobutane = 3 g

Now 7g isobutene is converted to isobutane by reduction to give $\frac{58 \times 7}{56}$ g

= 7.25 g of isobutane
Total mass of isobutene is 7.25+3 = 10.25 gm
Value of (a-0.25)/5 = (10.25-0.25)/5 = 2

FIITJEE REVIEW TEST-05

(Batches: 921 Lot)

IIT – JEE, 2021

MATHEMATICS

Time: 1 Hour **Maximum Marks: 59**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-04)** contains **04 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (05-09)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-A – (10-13)** contains **02 paragraphs**. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-C – (14-18)** contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks: – 2 In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part A (4),** you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **Session :**. . . . . **Date:**. . . . . .

SECTION : MATHEMATICS

PART – A: (Single Correct Answer Type)

*This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. The degree and order of differential equation of family of circle touching a parabola $y^2 = 4\lambda x$ ($\lambda \in R$) at its vertex is
   (A) degree 2, order 1 (B) degree 1, order 2
   (C) degree 1, order 0 (D) degree 1, order 1

2. Find the area of the region enclosed between the curve $y = 2x^4 - x^2$, the x-axis and the ordinates of the points where the curve has local minima
   (A) $\frac{7}{60}$ (B) $\frac{7}{80}$
   (C) $\frac{7}{120}$ (D) $\frac{7}{160}$

3. Area bounded by the curve $y = 1 - 2^{1+\sin x}$ and x-axis, $x \in \left[0, \frac{\pi}{2}\right]$ where $\int_0^{\pi/2} 2^{\sin x} \cdot dx = k$
   (A) $\frac{\pi - 4k}{2}$ (B) $\frac{\pi - 2k}{2}$
   (C) $\frac{2\pi - k}{2}$ (D) $\frac{4k - \pi}{2}$

4. The area bounded by $y = \cos^{-1}(\cos x)$, x –axis, $0 \le x \le 2\pi$, is
   (A) π (B) 2π
   (C) π^2 (D) $2\pi^2$

Space for rough work

PART – A: (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

5. The solution of $\left(\frac{dy}{dx}\right)^2 - 2\left(x + \frac{1}{4x}\right)\frac{dy}{dx} + 1 = 0$

(A) $y = x^2 + c$ (B) $y = \frac{1}{2}\ln(x) + c,\ x > 0$

(C) $y = \frac{x}{2} + c$ (D) $y = \frac{x^2}{2} + c$

6. Solution of differential equation $(e^x+1)\, y\, dy = (y+1)\, e^x\, dx$ is

(A) $(e^x + 1)(y+1) = ce^y$ (B) $(e^x+1)|(y+1)| = ce^y$

(C) $(e^x+1)(y+1) = \pm ce^y$ (D) None of these

7. If $2xydy = \left(x^2 + y^2 + 1\right)dx$, $y(1) = 0$ and $y(\alpha) = \sqrt{3}$ then α can be

(A) 2 (B) –2

(C) 5 (D) –5

8. Solution of $(1 - xy - x^5y^5)dx = x^2(x^4y^4 + 1)dy$ is

(A) $\log_e x = xy + \frac{(xy)^5}{5} + c$ (B) $\log_e\left(\frac{x}{e^{xy}}\right) = \frac{(xy)^5}{5} + c$

(C) $\log_e y = xy + \frac{(xy)^4}{4} + c$ (D) $\log_e\left(\frac{y}{e^{xy}}\right) = \frac{(xy)^5}{5} + c$

9. Area of the region bounded by the curve y = tanx and lines y = 0 and x = 1 is

(A) $\int_0^1 tan(1-x)\,dx$ (B) $\tan 1 - \int_0^{\tan 1} tan^{-1}x\,dx$

(C) $\int_0^{\tan 1} tan^{-1}y\,dy$ (D) $\int_0^1 tan^{-1}x\,dx$

Space for rough work

PART – A: (Paragraph Type)

*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 10 to 11

Suppose 'f' and 'g' are differentiable functions such that $xg[f(x)]\, f'[g(x)]\, g'(x) = f(g(x))\, g'(f(x))\, f'(x)\ \forall\ x \in R$ that 'f' is +ve, g is +ve $\forall\ x \in R$. Also $\int_0^x f\big(g(t)\big)\,dt = \frac{1}{2}\left(1-e^{-2x}\right) \forall\ x \in R$, g[f(0)] = 1, $h(x) = \frac{g\left[f(x)\right]}{f\left[g(x)\right]}\ \forall\ x \in R$.

10. The graph of y = h(x) is symmetric with respect to line
(A) x = – 1 (B) x = 0
(C) x = 1 (D) x = 2

11. The value of f(g(0)) + g(f(0)) =
(A) 1 (B) 2
(C) 3 (D) 4

Paragraph for Question Nos. 12 to 13

Consider function $f(x) = \sqrt{4-x^2}$, g(x) = |x – 2| and $h(x) = \sqrt{2-x}$, for $x \in R$ a function is defined as F(x) = max or min{f(x), g(x), h(x)} then

12. Area of F(x) = min{f(x), g(x), h(x)} between the co-ordinate axes for x < 0 is
(A) 2π sq. units (B) π sq. units
(C) 4π sq. units (D) none of these

13. Area enclosed by F(x) = min{f(x), g(x), h(x)}, F(x) = max{f(x), g(x), h(x)}, x = 0, y = 0 and x = 2 is
(A) π sq. units (B) $(\pi + 2)$ sq. units
(C) $\pi - \left(\frac{1-2^{7/2}}{6}\right)$ sq. units (D) $\pi + \left(\frac{1-2^{7/2}}{6}\right)$ sq. units

Space for rough work

PART – C: (Integer Answer Type)

*This section contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.*

14. The solution of $x^2dy - y^2dx + xy^2(x - y)dy = 0$ is $\ln\left|\frac{x-y}{xy}\right| = \frac{y^k}{2} + c$, then the value of k is _____

15. If y = f(x) is the solution of the differential equation $x(y^3 - x)dy = y(x + y^3)dx$ and $f(1) = (-2)^{1/3}$ and $f^{-1}(-2) = k$. Then |k| = _____ (Where|.| denotes the modulus function)

16. Area bounded by the curve $y = x^3 (x-1)^2$, y – axis and the line y = 8 is A then [A]/2 is _____ (where [.] denotes GIF)

17. Area enclosed by y = g(x), x = – 3, x = 5 and x–axis where g(x) is inverse function of $f(x) = x^3 + 3x + 1$ is A, then [A] is _____. (where [.] denotes the greatest integer function

18. The area enclosed by the curve max {|x – 1|, |y|} = k is 100, then k is equal to _____

Space for rough work

FIITJEE REVIEW TEST

MATHEMATICS

ANSWERS

MATHEMATICS (SECTION –III)

PART – A

1. **D**	2. **C**	3. **D**	4. **C**
5. **A, B**	6. **B, C**	7. **A, B**	8. **A, B**
9. **A, B**	10. **C**	11. **B**	12. **D**
13. **D**			

PART – C

14. **2**	15. **4**	16. **7**	17. **4**
18. **5**			

HINTS & SOLUTIONS

MATHEMATICS (SECTION –III)

PART – A

1. **D**

Clearly circle will be $(x + r)^2 + y^2 = r^2$; on differentiating both sides, we get

$2(x + r) + 2y\frac{dy}{dx} = 0 \Rightarrow r = -y\frac{dy}{dx} - x$

substituting in $(x + r)^2 + y^2 = r^2$

$\Rightarrow y^2\left(\frac{dy}{dx}\right)^2 + y^2 = \left(x + y\frac{dy}{dx}\right)^2$

$\Rightarrow 2xy\frac{dy}{dx} = y^2 - x^2 \Rightarrow$ order = 1, degree = 1.

2. **C**

$y = 2x^4 - x^2$

$y' = 8x^3 - 2x = 0 \Rightarrow x = 0, \frac{1}{2}, -\frac{1}{2}$

$y'' = 24x^2 - 2$

curve has local minim at $-\frac{1}{2}$ and $\frac{1}{2}$

$\therefore$ area $= \left|\int_{-1/2}^{1/2}(2x^4 - x^2)\,dx\right| = \frac{7}{120}$.

3. **D**

Area $= \left|\int_0^{\pi/2}(1 - 2^{1+\sin x})\,dx\right| = \left|\frac{\pi}{2} - 2\int_0^{\pi/2}2^{\sin x}\,dx\right| = \left|\frac{\pi}{2} - 2k\right| = \frac{4k - \pi}{2}$

4. **C**

$Y = \cos^{-1}(\cos x) = \begin{cases} x, & 0 \le x \le \pi \\ 2\pi - x & \pi < x \le 2\pi \end{cases}$

The required area $= \int_0^{\pi} x\,dx + \int_{\pi}^{2\pi}(2\pi - x)\,dx = \pi^2$

5. **A, B**

$\left(\frac{dy}{dx} - 2x\right)\left(\frac{dy}{dx} - \frac{1}{2x}\right) = 0$

$\therefore$ dy = 2x dx and $dy = \frac{1}{2x}dx$

$\Rightarrow y = x^2 + c$ and $y = \frac{1}{2}\ln x + c$, x > 0.

6. **B, C**

$(e^x+1)\,y\,dy = (y+1)e^x\,dx$

$\Rightarrow \frac{e^x}{e^x + 1}dx = \frac{y}{y + 1}dy \Rightarrow \ln(e^x+1) = y - \ln|y+1| + \ln c$

$\Rightarrow (e^x + 1)\,|(y+1)| = ce^y$

which can also be written as
$(e^x + 1)(y+1) = \pm ce^y$

7. **A, B**

We have $2xy\,dy = (x^2 + y^2 + 1)dx$

$$\frac{2xydy - y^2dx}{x^2} = \left(1 + \frac{1}{x^2}\right)dx$$

$$d\left(\frac{y^2}{x}\right) = d\left(x - \frac{1}{x}\right)$$

$$\frac{y^2}{x} = x - \frac{1}{x} + c$$

$y(1) = 0 \Rightarrow c = 0$

$y^2 = x^2 - 1$

$y(\alpha) = \sqrt{3}$

$\alpha^2 - 1 = 3$

$\alpha^2 = 4$

$\alpha = \pm 2$

8. **A, B**

$(1 - xy - x^5y^5)\,dx = (x^6y^4 + x^2)\,dy$

$(1 - xy)\,dx - x^2\,dy = x^6y^4dy + x^5y^5\,dx$

$(1 - xy)\,dx - x^2\,dy = x^5y^4\,(x\,dy + y\,dx)$

$dx - x(y\,dx + x\,dy) = x^5\,y^4\,[d(xy)]$

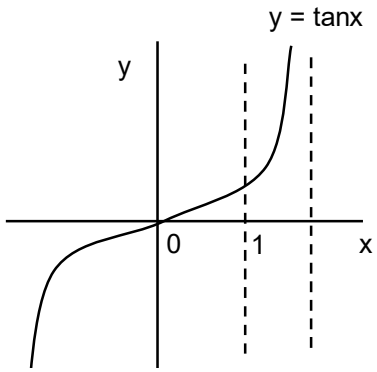
$$\int \frac{dx}{x} = \int \left(x^4y^4 + 1\right)\left(d(xy)\right) + c$$

$$\log_e x = xy + \frac{(xy)^5}{5} + c$$

9. **A, B**

Required area = $\int_0^1 \tan x\,dx = \int_0^1 \tan(1 - x)\,dx$

Also required area = $\tan 1 - \int_0^{\tan 1} \tan^{-1} y\,dy$.



10. **C**

11. **B**

10–11. $f[g(x)] = e^{-2x}$

$$\Rightarrow \quad \frac{x \cdot \left[f\left[g(x)\right]\right]'}{f\left[g(x)\right]} = \frac{\left[g\left[f(x)\right]\right]'}{g\left[f(x)\right]}$$

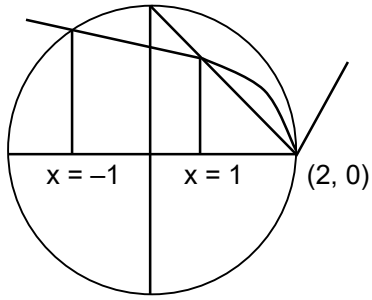
$\Rightarrow$ $g[f(x)] = e^{-x^2}$

$H(x) = e^{-(x-1)^2+1}$

(2) f[g(0)] + g[f(0)] = 2
(3) $H(x)_{max} = e$.

12. **D**

$y = \sqrt{4 - x^2}$
$y = |x - 2|$
$y = \sqrt{2 - x}$



The required area $= \int_{-2}^{-1} \sqrt{4 - x^2} + \int_{-1}^{0} \sqrt{2 - x}$

13. **D**

The required area

$$= \frac{\pi(2)^2}{4} - \int_0^1 \sqrt{2 - x}\, dx - \frac{1}{2}(1)(1)$$

$$= \pi + \frac{1 - 2^{7/2}}{6}$$

PART – C

14. **2**

$$x^2y^2\left(\frac{dy}{y^2} - \frac{dx}{x^2}\right) + x^2y^3\left(\frac{1}{y} - \frac{1}{x}\right)dy = 0$$

$$\Rightarrow d\left(\frac{1}{x} - \frac{1}{y}\right) + y\left(\frac{1}{y} - \frac{1}{x}\right)dy = 0$$

$$\Rightarrow \frac{d\left(\frac{1}{x} - \frac{1}{y}\right)}{\frac{1}{x} - \frac{1}{y}} = ydy$$

$$\Rightarrow \ln\left|\frac{1}{x} - \frac{1}{y}\right| = \frac{y^2}{2} + c \Rightarrow k = 2.$$

15. **4**

$x^2dy + xydx = y^3(xdy - ydx)$
$\Rightarrow xd(xy) = y^3x^2d(y/x)$

$$\Rightarrow \frac{d(xy)}{x^2y^2} = \frac{yd(\frac{y}{x})}{x}$$

$$\Rightarrow \frac{y^2}{2x^2} + \frac{1}{xy} = C \text{ (C is an integration constant)}$$

Put $x = 1 \Rightarrow C = \frac{y^3 + 2}{2y} = 0$

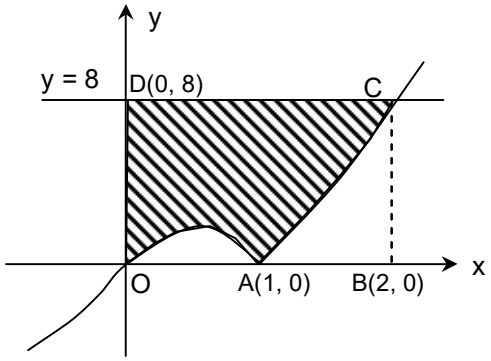
$\Rightarrow y^3 = -2x$ is the function.
$\Rightarrow -8 = -2x \Rightarrow x = 4$

16. **7**

Area = Area (OACDO)

$$\int_0^2 8-\left(x^5-2x^4+x^3\right)dx$$

$$= 16 - \frac{28}{15} = \frac{212}{15} Sq.units$$



17. **4**

$$A = \left(4-\int_0^1 \left(f(y)-1\right)dy\right)+\left|4-\int_{-1}^0 \left(1-f(y)\right)dy\right| = \frac{9}{2}.$$

18. **5**

Area enclosed by the region

ABCD = 2k · 2k

$\Rightarrow 4k^2 = 100$

k = 5

